MAT4305 Oblig

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March 2026

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1 Problem 1

1.1 Problem 1a

In this problem we need to show two things. First, we need to show that if $u \in L^2(\mathbb{R}^n)$, then

$$u \in H^k(\mathbb{R}^n)$$

if and only if

$$\omega(\xi) = (1 + |\xi|^2)^{k/2}\hat{u}(\xi)$$

belongs to $L^2(\mathbb{R}^n)$. Secondly, we need to show that the map

$$u \mapsto \|\omega\|_{L^2(\mathbb{R}^n)}$$

defines a norm on $H^k(\mathbb{R}^n)$, and that this norm is equivalent to the usual H^k-norm.

By the definition of $H^k(\mathbb{R}^n)$, we have

$$u \in H^k(\mathbb{R}^n)$$

if and only if

$$D^\alpha u \in L^2(\mathbb{R}^n)$$

for all multi-indices α with $|\alpha| \leq k$. Therefore,

$$\|u\|^2_{H^k(\mathbb{R}^n)} = \sum_{|\alpha|\leq k} \|D^\alpha u\|^2_{L^2(\mathbb{R}^n)}.$$

By Plancherel's theorem and the Fourier transform identity

$$\widehat{D^\alpha u}(\xi) = (i\xi)^\alpha \hat{u}(\xi),$$

we get

$$\|D^\alpha u\|^2_{L^2(\mathbb{R}^n)} = \int_{\mathbb{R}^n} |(i\xi)^\alpha \hat{u}(\xi)|^2 \, d\xi.$$

Since $|i| = 1$, this becomes

$$\|D^\alpha u\|^2_{L^2(\mathbb{R}^n)} = \int_{\mathbb{R}^n} |\xi^\alpha|^2 |\hat{u}(\xi)|^2 \, d\xi.$$

Taking the sum over all $|\alpha| \leq k$, we obtain

$$\sum_{|\alpha|\leq k} \|D^\alpha u\|^2_{L^2(\mathbb{R}^n)} = \sum_{|\alpha|\leq k} \int_{\mathbb{R}^n} |\xi^\alpha|^2 |\hat{u}(\xi)|^2 \, d\xi.$$

Since there are only finitely many multi-indices with $|\alpha| \leq k$, we may interchange the finite sum and the integral. Hence

$$\|u\|^2_{H^k(\mathbb{R}^n)} = \int_{\mathbb{R}^n} \left(\sum_{|\alpha|\leq k} |\xi^\alpha|^2 \right) |\hat{u}(\xi)|^2 \, d\xi.$$

Now we compare the two weights

$$\sum_{|\alpha|\le k} |\xi^\alpha|^2$$

and

$$(1+|\xi|^2)^k.$$

Since k is fixed, there exist constants $c, C > 0$ such that for all $\xi \in \mathbb{R}^n$,

$$c(1+|\xi|^2)^k \le \sum_{|\alpha|\le k} |\xi^\alpha|^2 \le C(1+|\xi|^2)^k.$$

Therefore,

$$\|u\|^2_{H^k(\mathbb{R}^n)} < \infty$$

if and only if

$$\int_{\mathbb{R}^n} (1+|\xi|^2)^k |\hat{u}(\xi)|^2 \, d\xi < \infty.$$

But

$$\int_{\mathbb{R}^n} (1+|\xi|^2)^k |\hat{u}(\xi)|^2 \, d\xi = \int_{\mathbb{R}^n} \left|(1+|\xi|^2)^{k/2}\hat{u}(\xi)\right|^2 \, d\xi = \|\omega\|^2_{L^2(\mathbb{R}^n)}.$$

Thus

$$u \in H^k(\mathbb{R}^n)$$

if and only if

$$\omega \in L^2(\mathbb{R}^n).$$

Next, we show that

$$\|u\|_* := \|\omega\|_{L^2(\mathbb{R}^n)}$$

defines a norm on $H^k(\mathbb{R}^n)$. The positivity and homogeneity follow directly from the corresponding properties of the L^2-norm. If

$$\|u\|_* = 0,$$

then

$$(1+|\xi|^2)^{k/2}\hat{u}(\xi) = 0$$

almost everywhere. Since the weight $(1+|\xi|^2)^{k/2}$ is strictly positive, we get

$$\hat{u}(\xi) = 0$$

almost everywhere. Hence $u = 0$ in $L^2(\mathbb{R}^n)$. The triangle inequality follows from the linearity of the Fourier transform and the triangle inequality in L^2.

Finally, by the comparison of weights above, there exist constants $c, C > 0$ such that

$$c\|\omega\|^2_{L^2(\mathbb{R}^n)} \le \|u\|^2_{H^k(\mathbb{R}^n)} \le C\|\omega\|^2_{L^2(\mathbb{R}^n)}.$$

Equivalently, after changing the constants, we obtain

$$c\|u\|_{H^k(\mathbb{R}^n)} \le \|\omega\|_{L^2(\mathbb{R}^n)} \le C\|u\|_{H^k(\mathbb{R}^n)}.$$

Therefore, the norm

$$u \mapsto \|\omega\|_{L^2(\mathbb{R}^n)}$$

is equivalent to the usual $H^k(\mathbb{R}^n)$-norm. The proof is finished.

1.2 Problem 1b

In this problem we need to show that

$$\|v\|_{C^{0,\alpha}(\mathbb{R}^n)} \leq C\|v\|_{H^k(\mathbb{R}^n)}$$

for

$$0 < \alpha < \min\left(1, k - \frac{n}{2}\right)$$

and for all $v \in H^k(\mathbb{R}^n)$.

The aim is that we first use the inverse Fourier transform to write $v(x)$. Then we control the L^∞-norm of v. Finally, we control the Hölder seminorm of v.

First, by the inverse Fourier transform, we have

$$v(x) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{ix\cdot\xi} \widehat{v}(\xi)\, d\xi.$$

Hence

$$|v(x)| \leq C \int_{\mathbb{R}^n} |\widehat{v}(\xi)|\, d\xi.$$

Now we apply the weight

$$(1 + |\xi|^2)^{-k/2}(1 + |\xi|^2)^{k/2}.$$

Then

$$\int_{\mathbb{R}^n} |\widehat{v}(\xi)|\, d\xi = \int_{\mathbb{R}^n} (1 + |\xi|^2)^{-k/2}(1 + |\xi|^2)^{k/2} |\widehat{v}(\xi)|\, d\xi.$$

Using the Cauchy–Schwarz inequality, we get

$$\int_{\mathbb{R}^n} |\widehat{v}(\xi)|\, d\xi \leq \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^{-k}\, d\xi\right)^{1/2} \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^k |\widehat{v}(\xi)|^2\, d\xi\right)^{1/2}.$$

Since $k > n/2$, we know that

$$\int_{\mathbb{R}^n} (1 + |\xi|^2)^{-k}\, d\xi < \infty.$$

Moreover, by Problem 1a, we have

$$\left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^k |\widehat{v}(\xi)|^2\, d\xi\right)^{1/2} \leq C\|v\|_{H^k(\mathbb{R}^n)}.$$

Therefore,

$$\|v\|_{L^\infty(\mathbb{R}^n)} \leq C\|v\|_{H^k(\mathbb{R}^n)}.$$

Next we need to estimate the Hölder seminorm. For $x, y \in \mathbb{R}^n$, by the inverse Fourier transform we get

$$v(x) - v(y) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \left(e^{ix\cdot\xi} - e^{iy\cdot\xi}\right) \widehat{v}(\xi)\, d\xi.$$

Hence

$$|v(x) - v(y)| \leq C \int_{\mathbb{R}^n} \left| e^{ix\cdot\xi} - e^{iy\cdot\xi} \right| |\widehat{v}(\xi)| \, d\xi.$$

For $0 < \alpha \leq 1$, we have the estimate

$$\left| e^{ix\cdot\xi} - e^{iy\cdot\xi} \right| \leq C_\alpha |x-y|^\alpha |\xi|^\alpha.$$

Therefore,

$$|v(x) - v(y)| \leq C|x-y|^\alpha \int_{\mathbb{R}^n} |\xi|^\alpha |\widehat{v}(\xi)| \, d\xi.$$

Now we again apply the weight

$$(1+|\xi|^2)^{-k/2}(1+|\xi|^2)^{k/2}.$$

Then

$$\int_{\mathbb{R}^n} |\xi|^\alpha |\widehat{v}(\xi)| \, d\xi = \int_{\mathbb{R}^n} |\xi|^\alpha (1+|\xi|^2)^{-k/2}(1+|\xi|^2)^{k/2} |\widehat{v}(\xi)| \, d\xi.$$

Using the Cauchy–Schwarz inequality, we obtain

$$\int_{\mathbb{R}^n} |\xi|^\alpha |\widehat{v}(\xi)| \, d\xi \leq \left(\int_{\mathbb{R}^n} |\xi|^{2\alpha} (1+|\xi|^2)^{-k} \, d\xi \right)^{1/2} \left(\int_{\mathbb{R}^n} (1+|\xi|^2)^k |\widehat{v}(\xi)|^2 \, d\xi \right)^{1/2}.$$

The first integral is finite if

$$\alpha < k - \frac{n}{2}.$$

Indeed, for large $|\xi|$, we have

$$|\xi|^{2\alpha}(1+|\xi|^2)^{-k} \sim |\xi|^{2\alpha-2k}.$$

Thus in polar coordinates the integral behaves like

$$\int_1^\infty r^{2\alpha-2k+n-1} \, dr,$$

which is finite if

$$2\alpha - 2k + n < 0.$$

This is equivalent to

$$\alpha < k - \frac{n}{2}.$$

Therefore, for

$$0 < \alpha < \min\left(1, k - \frac{n}{2}\right),$$

we have

$$\int_{\mathbb{R}^n} |\xi|^{2\alpha}(1+|\xi|^2)^{-k} \, d\xi < \infty.$$

Using again the result from Problem 1a, we get

$$|v(x) - v(y)| \leq C|x - y|^{\alpha} \|v\|_{H^k(\mathbb{R}^n)}.$$

Hence

$$\frac{|v(x) - v(y)|}{|x - y|^{\alpha}} \leq C\|v\|_{H^k(\mathbb{R}^n)}.$$

Taking the supremum over all $x \neq y$, we obtain

$$[v]_{C^{0,\alpha}(\mathbb{R}^n)} \leq C\|v\|_{H^k(\mathbb{R}^n)}.$$

Finally, since

$$\|v\|_{C^{0,\alpha}(\mathbb{R}^n)} = \|v\|_{L^\infty(\mathbb{R}^n)} + [v]_{C^{0,\alpha}(\mathbb{R}^n)},$$

we get

$$\|v\|_{C^{0,\alpha}(\mathbb{R}^n)} \leq C\|v\|_{H^k(\mathbb{R}^n)}.$$

The proof is finished.

1.3 Problem 1c

First we assume that $U \subset \mathbb{R}^n$ is a bounded domain with $\partial U \in C^1$. We also assume that $u \in H^k(U)$, where $k > n/2$.

In this problem we are going to prove that

$$\|u\|_{C(\overline{U})} \leq C\|u\|_{H^k(U)}.$$

In this problem we apply the extension theorem and use the result from Problem 1b.

Since U is a bounded domain with C^1-boundary, there exists a bounded linear extension operator

$$E : H^k(U) \to H^k(\mathbb{R}^n)$$

such that

$$Eu = u \quad \text{a.e. on } U,$$

and

$$\|Eu\|_{H^k(\mathbb{R}^n)} \leq C\|u\|_{H^k(U)}.$$

By the result from Problem 1b, we let

$$v = Eu.$$

Then $v \in H^k(\mathbb{R}^n)$, and we get

$$\|v\|_{C^{0,\alpha}(\mathbb{R}^n)} \leq C\|v\|_{H^k(\mathbb{R}^n)}.$$

In particular,

$$\|v\|_{C^0(\mathbb{R}^n)} \leq C\|v\|_{H^k(\mathbb{R}^n)}.$$

Since $v = Eu$, we obtain

$$\|v\|_{C^0(\mathbb{R}^n)} \leq C\|Eu\|_{H^k(\mathbb{R}^n)} \leq C\|u\|_{H^k(U)}.$$

Now $v = Eu = u$ almost everywhere on U, and v is continuous on $\mathbb{R}^n$. Therefore the restriction of v to $\overline{U}$ gives the continuous representative of u. Hence

$$\|u\|_{C(\overline{U})} \leq \|v\|_{C^0(\mathbb{R}^n)} \leq C\|u\|_{H^k(U)}.$$

The proof is finished.

2 Problem 2

2.1 Problem 2a

In this problem we need to prove that, for all $x \neq 0$ and all $\xi \in \mathbb{R}^n$,

$$|\xi|^2 \leq A(x)\xi \cdot \xi \leq \alpha|\xi|^2.$$

By the definition of $A(x)$, we have

$$A(x)\xi \cdot \xi = \sum_{i,j=1}^{n} \left(\delta_{ij} + (\alpha - 1)\frac{x_i x_j}{|x|^2}\right)\xi_i \xi_j.$$

Therefore,

$$A(x)\xi \cdot \xi = |\xi|^2 + (\alpha - 1)\frac{1}{|x|^2}\sum_{i,j=1}^{n} x_i x_j \xi_i \xi_j.$$

Since

$$\sum_{i,j=1}^{n} x_i x_j \xi_i \xi_j = \left(\sum_{i=1}^{n} x_i \xi_i\right)^2 = (x \cdot \xi)^2,$$

we get

$$A(x)\xi \cdot \xi = |\xi|^2 + (\alpha - 1)\frac{(x \cdot \xi)^2}{|x|^2}.$$

Since $\alpha > 1$, the second term is non-negative. Hence

$$A(x)\xi \cdot \xi \geq |\xi|^2.$$

For the upper bound, by the Cauchy–Schwarz inequality,

$$|x \cdot \xi|^2 \leq |x|^2|\xi|^2.$$

Thus

$$\frac{(x \cdot \xi)^2}{|x|^2} \leq |\xi|^2.$$

Therefore,

$$A(x)\xi \cdot \xi \leq |\xi|^2 + (\alpha - 1)|\xi|^2 = \alpha|\xi|^2.$$

Hence

$$|\xi|^2 \leq A(x)\xi \cdot \xi \leq \alpha|\xi|^2.$$

This shows that the operator

$$Lu = \operatorname{div}(ADu)$$

is uniformly elliptic, with ellipticity constants 1 and α.

2.2 Problem 2b

In this problem we assume that $0 < \varepsilon < 1$ is a parameter. and we fix i and define a function $u = x_i|x|^{1-n-\epsilon}$. And we assume that

$$\alpha = \frac{n-1}{\epsilon(\epsilon + n - 2)}$$

Then u is a classical solution of $Lu = 0$ on the open set $\mathbb{R}^n \setminus [0]$.

Since we know that $Lu = div(ADu)$, thus compute this operator and show this equal to 0.

Assume $u = x_i|x|^{1-n-\epsilon}$ and fix m , let $r = |x|$, and $\gamma = 1 - n - \epsilon$then we have

$$u(x) = x_m r^\gamma$$

First we compute the first time derivative for any j apply the chain rule and we can get that :

$$\partial_j u = \partial_j\big(x_m r^\gamma\big) = \delta_{mj} r^\gamma + \gamma x_m x_j r^{\gamma-2}.$$

In the problem we know that

$$a_{ij} = \delta_{ij} + (\alpha - 1)\frac{x_i x_j}{r^2}.$$

thus we can get that

$$a_{ij}\partial_j u = \left(\delta_{ij} + (\alpha-1)\frac{x_i x_j}{r^2}\right)\left(\delta_{mj} r^\gamma + \gamma x_m x_j r^{\gamma-2}\right).$$

And we multiplicate δ_{ij} and we can get that:

$$\delta_{ij}\delta_{mj} r^\gamma = \delta_{im} r^\gamma, \qquad \delta_{ij}\,\gamma x_m x_j r^{\gamma-2} = \gamma x_m x_i r^{\gamma-2}.$$

Since we know that $x_j\delta_{mj} = x_m$ and multiplicate $(\alpha-1)\frac{x_i x_j}{r^2}$ and we can get that:

$$(\alpha-1)\frac{x_i x_j}{r^2}\,\delta_{mj} r^\gamma = (\alpha-1) x_i x_m r^{\gamma-2},$$

Moreover

$$(\alpha-1)\frac{x_i x_j}{r^2}\,\gamma x_m x_j r^{\gamma-2} = (\alpha-1)\gamma x_i x_m r^{\gamma-4}\sum_j x_j^2 = (\alpha-1)\gamma x_i x_m r^{\gamma-2}.$$

Take summary and we can get

$$a_{ij}\partial_j u = \delta_{im} r^\gamma + \big[\gamma + (\alpha-1) + (\alpha-1)\gamma\big] x_m x_i r^{\gamma-2}.$$

put it in parentheses

$$\gamma + (\alpha-1) + (\alpha-1)\gamma = \alpha(\gamma+1) - 1.$$

and we can get that

$$a_{ij}\partial_j u = \delta_{im} r^\gamma + \big(\alpha(\gamma+1) - 1\big) x_m x_i r^{\gamma-2}.$$

Next step is find the derivatives for i and we can get that Thus,

$$Lu = \partial_i(a_{ij}\partial_j u) = \partial_i(\delta_{im} r^\gamma) + \big(\alpha(\gamma+1) - 1\big)\partial_i(x_m x_i r^{\gamma-2}).$$

First, we compute the first term:

$$\partial_i(\delta_{im} r^\gamma) = \partial_m(r^\gamma) = \gamma x_m r^{\gamma-2}.$$

For the second term, we use

$$\partial_i(x_m x_i r^{\gamma-2}) = (\gamma + n - 1) x_m r^{\gamma-2}.$$

Therefore,

$$Lu = \Big[\gamma + \big(\alpha(\gamma+1) - 1\big)(\gamma + n - 1)\Big] x_m r^{\gamma-2}.$$

Now substitute

$$\gamma = 1 - n - \varepsilon.$$

Then

$$\gamma + 1 = 2 - n - \varepsilon, \qquad \gamma + n - 1 = -\varepsilon.$$

Hence,

$$Lu = \Big[1 - n - \varepsilon - \varepsilon\big(\alpha(2 - n - \varepsilon) - 1\big)\Big] x_m r^{-n-1-\varepsilon}.$$

After simplification, we obtain

$$Lu = \Big[\alpha\varepsilon(\varepsilon + n - 2) - (n-1)\Big] x_m r^{-n-1-\varepsilon}.$$

Therefore, it suffices to choose

$$\alpha = \frac{n-1}{\varepsilon(\varepsilon + n - 2)},$$

and then we obtain the desired result.

Thus we can get $Lu = 0$ in $\mathbb{R}^n \setminus 0$

2.3 Problem 2c

In this problem we need to show that the classical partial derivatives of u on $\mathbb{R}^n \setminus \{0\}$ are also weak derivatives of u on $\mathbb{R}^n$.

First we need to find the classical derivative of $u(x)$. This means that for any j we have

$$\partial_j u = \partial_j(x_i)\,|x|^{1-n-\varepsilon} + x_i\,\partial_j\big(|x|^{1-n-\varepsilon}\big).$$

Hence,

$$\partial_j u(x) = \delta_{ij}|x|^{1-n-\varepsilon} + x_i\,\partial_j\big(|x|^{1-n-\varepsilon}\big).$$

Because

$$\partial_j\big(|x|^m\big) = m|x|^{m-2}x_j,$$

taking $m = 1-n-\varepsilon$, we obtain

$$\partial_j u(x) = \delta_{ij}|x|^{1-n-\varepsilon} + (1-n-\varepsilon)x_i x_j|x|^{-n-\varepsilon-1}.$$

This is the classical derivative of u on $\mathbb{R}^n \setminus \{0\}$.

For the weak derivative to exist for u, we must show that u and $\partial_j u$ are locally integrable. That is, we need to show

$$u, \partial_j u \in L^1_{\text{loc}}(\mathbb{R}^n).$$

We consider the ball $B_1(0)$, and we have

$$|u(x)| = |x_i||x|^{1-n-\varepsilon} \leq |x|^{2-n-\varepsilon}.$$

Recall that $|x_i| \leq |x|$. Using polar coordinates, we get

$$\int_{B_1(0)} |u(x)|\,dx \leq \int_0^1 r^{2-n-\varepsilon} r^{n-1}\,dr = \int_0^1 r^{1-\varepsilon}\,dr < \infty.$$

The two terms in the expression for $\partial_j u(x)$ are both of order $|x|^{1-n-\varepsilon}$. Therefore,

$$|\partial_j u(x)| \lesssim |x|^{1-n-\varepsilon}.$$

It follows that

$$\int_{B_1(0)} |\partial_j u(x)|\,dx \leq \int_0^1 r^{1-n-\varepsilon} r^{n-1}\,dr = \int_0^1 r^{-\varepsilon}\,dr < \infty.$$

Hence,

$$u, \partial_j u \in L^1_{\text{loc}}(\mathbb{R}^n).$$

Next, we consider the integral on $\mathbb{R}^n \setminus 0$. This means that for any test function $\varphi \in C_c^\infty(\mathbb{R}^n)$ and for $\rho > 0$, we consider

$$D_\rho = \mathbb{R}^n \setminus \overline{B_\rho(0)}.$$

Since u is smooth on D_ρ, the classical derivative exists there, and thus we get

$$\int_{D_\rho} u\,\partial_j\varphi\,dx = -\int_{D_\rho} (\partial_j u)\,\varphi\,dx + \int_{\partial D_\rho} u\varphi\,\nu_j\,dS.$$

Moreover,

$$\partial D_\rho = \partial B_\rho(0), \qquad \nu = -\frac{x}{|x|},$$

and hence

$$\int_{\partial D_\rho} u\varphi\,\nu_j\,dS = -\int_{|x|=\rho} u(x)\varphi(x)\,\frac{x_j}{|x|}\,dS.$$

Now we estimate the boundary term. On $|x| = \rho$, we have

$$|u(x)| = |x_i||x|^{1-n-\varepsilon} \le \rho^{2-n-\varepsilon}.$$

Therefore,

$$\left|\int_{\partial D_\rho} u\varphi\,\nu_j\,dS\right| \le \|\varphi\|_{L^\infty}\int_{|x|=\rho} |u(x)|\,dS \lesssim \rho^{2-n-\varepsilon}\rho^{n-1} = \rho^{1-\varepsilon}.$$

Since $0 < \varepsilon < 1$, we have

$$\rho^{1-\varepsilon} \to 0 \qquad \text{as } \rho \to 0.$$

Hence,

$$\int_{\partial D_\rho} u\varphi\,\nu_j\,dS \to 0.$$

Since $u, \partial_j u \in L^1_{\text{loc}}(\mathbb{R}^n)$, we may let $\rho \to 0$ in the identity above and obtain

$$\int_{\mathbb{R}^n} u\,\partial_j\varphi\,dx = -\int_{\mathbb{R}^n} (\partial_j u)\,\varphi\,dx.$$

Therefore, the classical partial derivative $\partial_j u$ on $\mathbb{R}^n \setminus \{0\}$ is also the weak derivative of u on $\mathbb{R}^n$.

2.4 Problem 2d

In this problem we are going to prove $Lu = 0$ weakly on $\mathbb{R}^n$.

This means we need to prove

$$Lu = div(ADu) = 0$$

in the weak sence on $\mathbb{R}^n$.

Since we know that the classical derivatives of u on $\mathbb{R}^n$ is also a weak solution.

We consider any test function $\phi \in C_c^\infty$ and consider the ball

$$D_\rho = \mathbb{R}^n \setminus \overline{B_\rho(0)}.$$

By the problem b and u is smooth on D_ρ Then we have

$Lu = div(ADu) = 0$ then we can take the integral by parts for this, this means

$$0 = -\int_{D_\rho} ADu \cdot D\phi dx + \int_{\partial D_\rho} \phi ADu \cdot \nu$$

thus we can get that :

$$\int_{D_\rho} ADu \cdot D\phi dx = \int_{\partial D_\rho} \phi ADu \cdot \nu$$

We now show that the boundary term tends to zero as $\rho \to 0$.

Recall from part b that, if $u = x_m r^\gamma$, where

$$\gamma = 1 - n - \varepsilon,$$

then

$$ADu = \left(\delta_{im} r^\gamma + \left(\alpha(\gamma+1) - 1\right) x_m x_i r^{\gamma-2}\right)_{i=1}^n.$$

On $\partial D_\rho = \partial B_\rho(0)$, the outward unit normal of D_ρ is

$$\nu = -\frac{x}{|x|}.$$

Hence

$$ADu \cdot \nu = -\alpha(\gamma+1) x_m r^{\gamma-1}.$$

Therefore,

$$\int_{\partial D_\rho} \phi ADu \cdot \nu \, dS = -\alpha(\gamma+1)\rho^{\gamma-1} \int_{|x|=\rho} \phi(x) x_m \, dS.$$

Since $\phi \in C_c^\infty(\mathbb{R}^n)$, we have

$$\phi(x) = \phi(0) + O(|x|) \quad \text{as } x \to 0.$$

Also, by symmetry,

$$\int_{|x|=\rho} x_m \, dS = 0.$$

Thus

$$\int_{|x|=\rho} \phi(x) x_m \, dS = \int_{|x|=\rho} (\phi(x) - \phi(0)) x_m \, dS.$$

Hence

$$\left| \int_{|x|=\rho} \phi(x) x_m \, dS \right| \le C\rho \int_{|x|=\rho} |x_m| \, dS \le C\rho \cdot \rho \cdot \rho^{n-1} = C\rho^{n+1}.$$

Since

$$\gamma - 1 = -n - \varepsilon,$$

we obtain

$$\left| \int_{\partial D_\rho} \phi A Du \cdot \nu \, dS \right| \leq C\rho^{-n-\varepsilon}\rho^{n+1} = C\rho^{1-\varepsilon}.$$

Because $0 < \varepsilon < 1$, this tends to zero as $\rho \to 0$.

Therefore, letting $\rho \to 0$, we get

$$\int_{\mathbb{R}^n} ADu \cdot D\phi \, dx = 0$$

for all $\phi \in C_c^\infty(\mathbb{R}^n)$. Hence

$$Lu = \operatorname{div}(ADu) = 0$$

in the weak sense on $\mathbb{R}^n$.

2.5 Problem 2e

In this problem we show that

$$u \in W^{1,\beta}(B(0,1)) \quad \Longleftrightarrow \quad \beta < \frac{n}{n+\varepsilon-1}.$$

Recall that

$$u(x) = x_i|x|^{1-n-\varepsilon}.$$

Let

$$B_1 = B(0,1).$$

We need to study the integrability of u and Du near the origin, since away from the origin the function is smooth.

First, we estimate u. Since

$$|x_i| \leq |x|,$$

we have

$$|u(x)| = |x_i||x|^{1-n-\varepsilon} \leq |x|^{2-n-\varepsilon}.$$

Therefore,

$$|u(x)|^\beta \leq |x|^{\beta(2-n-\varepsilon)}.$$

Using polar coordinates, we get

$$\int_{B_1} |u(x)|^\beta \, dx \leq C \int_0^1 r^{\beta(2-n-\varepsilon)} r^{n-1} \, dr.$$

This integral is finite if

$$\beta(2-n-\varepsilon) + n - 1 > -1.$$

Equivalently,

$$\beta(2-n-\varepsilon) + n > 0.$$

Since

$$2 - n - \varepsilon = -(n + \varepsilon - 2),$$

this gives

$$\beta < \frac{n}{n + \varepsilon - 2}.$$

Hence

$$u \in L^{\beta}(B_1)$$

whenever

$$\beta < \frac{n}{n + \varepsilon - 2}.$$

Next, we estimate the gradient of u. For each $j = 1, \ldots, n$, we compute

$$\partial_j u = \partial_j \left(x_i |x|^{1-n-\varepsilon} \right).$$

By the product rule,

$$\partial_j u = \delta_{ij} |x|^{1-n-\varepsilon} + x_i \partial_j \left(|x|^{1-n-\varepsilon} \right).$$

Since

$$\partial_j (|x|^m) = m|x|^{m-2} x_j,$$

with

$$m = 1 - n - \varepsilon,$$

we obtain

$$\partial_j u = \delta_{ij} |x|^{1-n-\varepsilon} + (1 - n - \varepsilon) x_i x_j |x|^{-n-\varepsilon-1}.$$

Since

$$|x_i x_j| \leq |x|^2,$$

we get

$$|\partial_j u| \leq C|x|^{1-n-\varepsilon}.$$

Thus

$$|Du(x)| \leq C|x|^{1-n-\varepsilon}.$$

Therefore,

$$|Du(x)|^{\beta} \leq C|x|^{\beta(1-n-\varepsilon)}.$$

Again using polar coordinates,

$$\int_{B_1} |Du(x)|^{\beta} \, dx \leq C \int_0^1 r^{\beta(1-n-\varepsilon)} r^{n-1} \, dr.$$

This integral is finite if

$$\beta(1 - n - \varepsilon) + n - 1 > -1.$$

Equivalently,

$$\beta(1 - n - \varepsilon) + n > 0.$$

Since

$$1 - n - \varepsilon = -(n + \varepsilon - 1),$$

this is equivalent to

$$\beta < \frac{n}{n + \varepsilon - 1}.$$

Hence

$$Du \in L^{\beta}(B_1)$$

whenever

$$\beta < \frac{n}{n + \varepsilon - 1}.$$

Now observe that

$$n + \varepsilon - 1 > n + \varepsilon - 2,$$

and hence

$$\frac{n}{n + \varepsilon - 1} < \frac{n}{n + \varepsilon - 2}.$$

Therefore the condition coming from the gradient is stronger than the condition coming from u itself. Thus, if

$$\beta < \frac{n}{n + \varepsilon - 1},$$

then both

$$u \in L^{\beta}(B_1)$$

and

$$Du \in L^{\beta}(B_1).$$

Hence

$$u \in W^{1,\beta}(B_1).$$

It remains to prove the converse. We show that if

$$\beta \geq \frac{n}{n + \varepsilon - 1},$$

then

$$Du \notin L^{\beta}(B_1).$$

Write $x = r\theta$, where $r = |x|$ and $\theta \in S^{n-1}$. From the formula above, we can write

$$\partial_j u(x) = r^{1-n-\varepsilon} \left(\delta_{ij} + (1 - n - \varepsilon)\theta_i \theta_j \right).$$

In particular, the angular factor is not identically zero. For example, at $\theta = e_i$, we have

$$\partial_i u = r^{1-n-\varepsilon} \left(1 + 1 - n - \varepsilon \right) = (2 - n - \varepsilon) r^{1-n-\varepsilon},$$

and

$$2 - n - \varepsilon \neq 0$$

for $n \geq 2$ and $0 < \varepsilon < 1$. Hence there exists a small cone $\Gamma \subset B_1$ around the x_i-axis and a constant $c > 0$ such that

$$|Du(x)| \geq c|x|^{1-n-\varepsilon}$$

for all $x \in \Gamma$.

Therefore,

$$\int_{B_1} |Du(x)|^\beta \, dx \geq \int_\Gamma |Du(x)|^\beta \, dx \geq c \int_0^1 r^{\beta(1-n-\varepsilon)} r^{n-1} \, dr.$$

The last integral diverges if

$$\beta(1-n-\varepsilon) + n - 1 \leq -1.$$

Equivalently,

$$\beta(1-n-\varepsilon) + n \leq 0.$$

This is the same as

$$\beta \geq \frac{n}{n+\varepsilon-1}.$$

Thus, if

$$\beta \geq \frac{n}{n+\varepsilon-1},$$

then

$$Du \notin L^\beta(B_1),$$

and therefore

$$u \notin W^{1,\beta}(B_1).$$

Combining both directions, we conclude that

$$u \in W^{1,\beta}(B(0,1)) \quad \Longleftrightarrow \quad \beta < \frac{n}{n+\varepsilon-1}.$$

The proof is finished.

2.6 Problem 2f

Let

$$B = B(0,1).$$

We consider the equation

$$Lv = \operatorname{div}(ADv) = 0$$

in B, with boundary condition

$$v|_{\partial B} = u|_{\partial B}.$$

On ∂B, we have $|x| = 1$. Since

$$u(x) = x_i |x|^{1-n-\varepsilon},$$

it follows that

$$u|_{\partial B} = x_i.$$

Therefore the boundary condition is

$$v|_{\partial B} = x_i.$$

Choose the smooth function

$$w(x) = x_i.$$

Then $w \in C^\infty(\overline{B})$, and

$$w|_{\partial B} = x_i = u|_{\partial B}.$$

We now write

$$v = w + z,$$

where $z \in H_0^1(B)$. Then v satisfies the boundary condition in the trace sense.

The weak formulation of

$$\operatorname{div}(ADv) = 0$$

is

$$\int_B ADv \cdot D\phi \, dx = 0$$

for all $\phi \in H_0^1(B)$. Since $v = w + z$, this becomes

$$\int_B AD(w + z) \cdot D\phi \, dx = 0.$$

Equivalently,

$$\int_B ADz \cdot D\phi \, dx = -\int_B ADw \cdot D\phi \, dx$$

for all $\phi \in H_0^1(B)$.

We define the bilinear form

$$a(z, \phi) = \int_B ADz \cdot D\phi \, dx$$

on $H_0^1(B)$, and the linear functional

$$F(\phi) = -\int_B ADw \cdot D\phi \, dx.$$

Then we want to find $z \in H_0^1(B)$ such that

$$a(z, \phi) = F(\phi)$$

for all $\phi \in H_0^1(B)$.

First, we check that a is bounded. Since A is bounded, we have

$$|A(x)\eta| \leq C|\eta|$$

for all $\eta \in \mathbb{R}^n$. Hence

$$|a(z,\phi)| \leq C \int_B |Dz||D\phi|\, dx.$$

By the Cauchy–Schwarz inequality,

$$|a(z,\phi)| \leq C\|Dz\|_{L^2(B)}\|D\phi\|_{L^2(B)} \leq C\|z\|_{H^1(B)}\|\phi\|_{H^1(B)}.$$

Thus a is bounded.

Next, we check coercivity. By the ellipticity estimate from Problem 2a,

$$A(x)\eta \cdot \eta \geq |\eta|^2.$$

Taking $\eta = Dz(x)$, we get

$$a(z,z) = \int_B ADz \cdot Dz\, dx \geq \int_B |Dz|^2\, dx = \|Dz\|^2_{L^2(B)}.$$

Since $z \in H^1_0(B)$, the Poincaré inequality gives

$$\|z\|_{L^2(B)} \leq C\|Dz\|_{L^2(B)}.$$

Therefore,

$$\|z\|^2_{H^1(B)} = \|z\|^2_{L^2(B)} + \|Dz\|^2_{L^2(B)} \leq C\|Dz\|^2_{L^2(B)}.$$

Hence

$$a(z,z) \geq c\|z\|^2_{H^1(B)}.$$

So a is coercive on $H^1_0(B)$.

It remains to check that F is bounded. Since

$$w(x) = x_i,$$

we have

$$Dw = e_i,$$

which is a constant vector. Hence $Dw \in L^2(B)$. Since A is bounded, we obtain

$$|F(\phi)| = \left| \int_B ADw \cdot D\phi\, dx \right| \leq C \int_B |Dw||D\phi|\, dx.$$

By the Cauchy–Schwarz inequality,

$$|F(\phi)| \leq C\|Dw\|_{L^2(B)}\|D\phi\|_{L^2(B)} \leq C\|\phi\|_{H^1(B)}.$$

Thus F is a bounded linear functional on $H^1_0(B)$.

By the Lax–Milgram theorem, there exists a unique

$$z \in H_0^1(B)$$

such that

$$a(z, \phi) = F(\phi)$$

for all $\phi \in H_0^1(B)$.

Now define

$$v = w + z.$$

Since $w \in H^1(B)$ and $z \in H_0^1(B)$, we have

$$v \in H^1(B).$$

Moreover,

$$v|_{\partial B} = w|_{\partial B} + z|_{\partial B} = x_i + 0 = x_i = u|_{\partial B}.$$

Also, by construction,

$$\int_B ADv \cdot D\phi \, dx = 0$$

for all $\phi \in H_0^1(B)$. Hence $v \in H^1(B)$ is a weak solution of

$$Lv = 0$$

with the required boundary condition.

The proof is finished.

2.7 Problem 2g

Now suppose that $n = 2$. By Problem 2e, we have

$$u \in W^{1,\beta}(B(0,1)) \quad \Longleftrightarrow \quad \beta < \frac{2}{1+\varepsilon}.$$

Since $\varepsilon > 0$, we have

$$\frac{2}{1+\varepsilon} < 2.$$

In particular,

$$u \notin W^{1,2}(B(0,1)) = H^1(B(0,1)).$$

By Problem 2f, there exists a weak solution

$$v_0 \in H^1(B(0,1))$$

of

$$Lv_0 = 0$$

with boundary condition

$$v_0|_{\partial B} = u|_{\partial B}.$$

On the other hand, by Problem 2d, u is a weak solution of

$$Lu = 0$$

on $B(0,1)$. Here the weak formulation is understood in the class $W^{1,\beta}(B(0,1))$, for every

$$1 < \beta < \frac{2}{1+\varepsilon}.$$

This is enough for the integrals in the weak formulation to be well-defined.

Now define

$$h = u - v_0.$$

Since u and v_0 have the same boundary value, we have

$$h|_{\partial B} = 0$$

in the trace sense. Also, by linearity,

$$Lh = Lu - Lv_0 = 0$$

weakly in B.

For each $t \in \mathbb{R}$, define

$$v_t = v_0 + th.$$

Equivalently,

$$v_t = (1-t)v_0 + tu.$$

Since the equation is linear, we get

$$Lv_t = 0$$

weakly in B. Moreover,

$$v_t|_{\partial B} = v_0|_{\partial B} + th|_{\partial B} = u|_{\partial B}.$$

Thus each v_t satisfies the required boundary condition.

Now choose

$$1 < \beta < \frac{2}{1+\varepsilon}.$$

Then

$$u \in W^{1,\beta}(B),$$

and since

$$v_0 \in H^1(B) = W^{1,2}(B),$$

we also have

$$v_0 \in W^{1,\beta}(B),$$

because B has finite measure and $\beta < 2$. Hence

$$v_t \in W^{1,\beta}(B)$$

for every $t \in \mathbb{R}$.

The solutions v_t are distinct. Indeed, if $u = v_0$, then since $v_0 \in H^1(B)$, we would have

$$u \in H^1(B),$$

which contradicts the fact that $u \notin H^1(B)$. Hence

$$h = u - v_0 \neq 0.$$

Therefore the family

$$\{v_t : t \in \mathbb{R}\}$$

contains infinitely many distinct weak solutions in $W^{1,\beta}(B)$.

Finally, when $n = 2$, the relation between α and ε becomes

$$\alpha = \frac{n-1}{\varepsilon(\varepsilon + n - 2)} = \frac{1}{\varepsilon^2}.$$

As $\varepsilon \to 0^+$, we have

$$\frac{2}{1+\varepsilon} \to 2.$$

Therefore, by choosing $\varepsilon > 0$ sufficiently small, equivalently by choosing α sufficiently large, the exponent $\beta < 2$ can be chosen arbitrarily close to 2.

The proof is finished.